

Active Inference ~ Twitter spaces #001 ~ December 9th 2021

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it might be just very lagged but then how can that happen anyways doesn't matter okay well all right it should be like recording i hope from this part on at least and uh yeah welcome thanks for suggesting to do it and just feel free to kind of start off however you want all right you know thanks uh thanks for hosting this and and and thank you for joining um so where did where do i start right like it's a i'm trying to attack a fairly fairly big topic here of connecting the essentially the eastern contemplative traditions with the very western scientific rational point of view you know the good start would be to i guess to explain really what buddhism is um and uh rather than what it appears to be right like ultimately buddhism is a technology of mind it simply is uh teaching about the nature of reality and um the whole structure that grew around it uh so that it became a religion is kind of a vehicle that was necessary to propagate the teaching through the ages you know it was a structure that was necessary to keep it keep it going keep it clear keep it reproduced over you know across generations and so on so forth and anybody who understands the teaching uh it's the key to to reality and this is the the the realization of a of a sort of like a buddha mind your your realization of your true uh nature and then you sort of get the keys to the kingdom you you have control over your own reality basically um any reality that you want to experience becomes a choice of the mind and it's a it's a very desirable state of mind and this is this is the state of mind that enlightened beings that um already are in and there are lots of them and you know countless countless buddhas and police at us but we don't see them because they don't exist in our in our realm and it's just a matter of essentially a perception and it's a change in perception as a result of a experience and as a result of knowledge and experience that's necessary in order to transition into you know the full capacity of mind which is simply that mind generates reality reality there's no there's no such thing as material reality um in from the buddhist perspective and i can argue that there is no such thing as the reality from the perspective of you know physics uh it's it's simply you know what reality is it's or what we experience as reality the qualia it's simply

what um the fundamental fabric of the universe which is the universe itself the the computational kind of fractal that the universe is you know appears to itself it has to appear like something so it appears as some kind of experience and what that appearance is also is kind of arbitrary so you know realization of the the buddha mind and you know entering nirvana is essentially a state of mind where when you experience at least visually it's something kind of like looking at an extremely high quality hdr video on an oled screen not you know 120 fps with full control of you creating that reality for yourself in just about as deep capacity as you want you know like you have infinite compute capacity to do this that's the um that's the kind of exp that's the kind of experience of an enlightened being you know from the buddhist perspective and the point of buddhism is essentially a teaching how to get there um it's it's it's about how to realize that an active inference and the free energy principle exactly the key uh to the kingdom that's that's exactly how we get there you know the free energy principle um there there are many many different ways just to summarize just including you know what it states but one useful quote that always comes to my mind is from frank herbert's dune uh it's a proverb of the ban na jesuit that be prepared to appreciate what you experience this is essentially you know kind of a meditation very contemplative meditative statement of of the free energy principle you have to be prepared to appreciate so you have to have a transcendental knowledge of some sort a knowledge of something that you might experience you may experience something that you have not experienced before you know you you put you feel a new um basis for um describing your experience an accumulation of these bases is simply how we create the model the generative model of the world um we we essentially you know accumulate bases in which we can express our experience then create a kind of a map of that that reality and then uh generate all kinds of possibilities out of that map and that gives us the new new directions of you know thinking but also it's a new direction of acting in the physical space and and so on and so forth this principle is just it just is completely universal that is essentially the essence of the buddhist teachings that you have to understand what is the fundamental reality and and why it is the way it is and um the argument there is really that you know the buddhist view of the world is correct simply because the world just cannot be any other way it's a consequence of the fact that any that something exists if you assume that something exists then you get the free energy principle out of it because if something exists this has to be separate from not the thing right like there has to be uh the a boundary then the mark of blanket and the mark of blanket um you know uh becomes essentially this fabric that creates uh categories of experience the things that we that we

actually experience as separate things but ultimately under the hood is just one universe the mind the buddha mind and that's that can be made mathematically precise the stephen wolfram at the end of 2019 he kicked off the world from physics project to find a fundamental theory of physics and he essentially put physics on top of the language of computation on hyper graphs rewrite rules on the hyper graph which is a very you know a computational model as good as any any other combinatory logic of lambda calculus all these all these things would be would be pretty good but for vultrium this particular forbidden gave him a really interesting insight into the structure of gravity and uh how that relates to quantum theory but basically what his model is saying is exactly that reality is entirely virtual it's not only that um you know there's no one physics there are infinitely many experiences of infinite possible physics in infinite number of universes that are co-present um and interact with each other in some sort of complex ways in such a way that um the resulting story as perceived by consciousness is within the universe the observing consciousness is not um you know causally inconsistent and the mark of blanket in active inference is basically you know um the description of what happens at the boundary between these these computational fractals you know as far as i was able to determine it is you know the most the most useful and most fundamental principle uh in in nature in terms of like you know actively um uh for actively acting in the world understanding what's going on uh acquiring knowledge and all of that like meditating on the principle itself is extremely useful as a as a tool for training the mind to sort of you know appreciate what is that we and um and this is where it gets really really interesting when you start thinking about like okay so what is out there and what is it that i actually experience so i can say for instance like okay i see um a nice and stable image of the world where um you know things have definite shapes and colors and um you know they appear solid and so on and so forth but how is my perception being encoded um you know i mean decoded from the signal that they actually see and we actually perceive a very small amount of our vision right this in color um most of it is black and white most of it is blurry and yet the the image that we see the image in the mind is a composite a result of a distributed computation um carried out by the by the brain right and it's um the brain is roughly composed of 150 000 uh functional uh units the cortical columns and they vote so to speak or produce some sort of average and that average is what we what we experience as world it's a world you know even as experienced by us is not something that's that's that's that's really definite we we don't have any way of definitely knowing what's out there on the other side of perception we can only say what it appears appears to be the the world that we actually are in you know it is the the this one thing

the the the void the the you know from hindu perspective will be the brahman the the the supreme self the uh from the hermetic perspective uh that would be the all uh from his first manchester's like four and a half thousand years ago he already described the the nature of reality he said that the universe is mental everything exists in the in the mind of the all that's you know that that is that is the other side so it's a it's a kind of like a co-recursive construction that the reality appears to be stable to us and appears to be um you know normal and ordinary just because we are all in the habit of being in that kind of state of mind but it only takes a different kind of state of mind that can be actually realized as a result of you know training and uh practice and meditation and visualizations and so on so forth to actually affect the reality around you it's it's possible and you know like um alexandra david nell for instance she was a a traveler um who spent i think about 50 years or or so in in tibet the the kind of things she described that people do there um you know were kind of extraordinary real magic and that's just something quite ordinary over there um at least you know until 1963 or whenever china took over um they developed that kind of technology um you know uh for a long time the bats was a was a place where in 12 cent uh i think it's 13th century the the roller of the bed married nine princesses from nine neighboring kingdoms and each of them brought uh with her um you know buddhist monks to import the technology into into the bat and then they essentially you know reorganize the entire country to be dedicated to study and and and and understanding the the all the consequences of of you know the the lesson of um uh siddhartha gautama so you know active inference um it's it's really it really is a way to get there like that meditating on this on this principle and putting it in in action and applying it in um you know um in a buddhist way of life is is a sure way to get to enlightenment it is not something that is you know within the realm of possibility to realize across you know in some so ever many lifetimes as it used to be the case you know buddhism um it is rather pessimistic worldview that because of your karmic burden you may not be able to get to enlightenment in this uh one lifetime but because now we have precise mathematical foundation to understand what is the optimal path you can actually do it you can you can just get there within a lifetime so um you know it's a i found it to be a fascinating system of fault and um it it starts to have effects on my life that you know um things do start to look a little bit science fiction it strange things are happening to me um like just you know luck just somehow finds me more and more that's the kind of situation so the path kind of gets easier as you keep going because the brain just automatically gives itself the work once you have the complete understanding of the world that comes with the buddhist viewpoint the brain eventually automatically and gears itself to

analyze all this information so that you know you start just living your life extreme level of confidence that you're always doing the right thing that you're always going the right way then you're always saying the right thing and so on so forth that um everything is as it should be it just becomes automatic it's a um you know one it's one of the meanings of the of nirvana or of liberation it doesn't mean that you um you know becomes uh mindless it's just like you you just know as a synthesis of your um of your understanding that what you're doing is simply the right course of action and it's a feeling that you you develop just like you know a reinforcement learner would have precise description of its signal of like how well it's doing while learning that's the kind of feedback loop um so build you know active inference is all about building all of that and essentially you know in some sense mathematically what what really means is that you want to become the market blanket that that is that is the name of the game you you just want to be the marker blanket the the thin veil between um you know two different virtual realities essentially your your own internal one and um you know the external world all right um i i think that's that's enough on the on the connection awesome thanks for sharing it and having it just in one direct place so i took a bunch of notes and there's a bunch of places to jump in but just what would be your goals or what would you like to do um you know i'd just like to talk about these ideas uh whatever if there's anything that you found interesting in this i i can expand on that um as much as you want uh basically there aren't that many people in the field who are interested in looking at um the world holistically this way that we have to integrate all the thinking in the western intellectual traditions with all the eastern contemplative traditions so i i like to talk about it from from that perspective and you know all i want to do is to just spread that idea that's that's why i'm here all right cool well in the youtube live chat where anyone can add a comment i'm also just kind of writing down some key themes and points to return back to we got the full first wash of paint on the mural and then in roughly chronological order here's a few points we could like return to hear a little bit more about your perspective and about different world knowledge traditions and then also connected to kind of the active basics so very early on you described buddhism as a technology and then like mind generating reality so the idea of mind generating conscious experience as well as the more sensory and less consciousness oriented take you provided on visual input which is like we're experiencing a generative visual model not a recognition sensory processing model from our retina the commonality there is this like generative model so i just wanted to hear a little more about the generative model or the generative perspective on consciousness how do we know what other what other possibilities are there is buddhism

unique here how do different buddhist groups think about this or what's the history on it that's a that's a that's a big question but generally um the way to look at analyzing the problem of consciousness is that every culture is correct within their context so uh buddhism is not it's not a superior point of view it's just an excellent one because it's been designed with a certain uh you know teaching tradition but there are a lot of metaphoric more metaphorical points of view that describe the fundamental nature of reality that are also simply correct and and they may appear uh contradictory like for instance you know um unification of all religions that essentially all the scriptures if you you really listen to the prophets they say the same thing um you know islam christianity judaism any major any major monotheistic religion just simply has to have the same kind of structure resulting from it and the variations really show in social structures um so you can think about these things as a what's going on is really with the global consciousness is that the universe is trying to understand itself and understand itself in such a way that it not only understands itself and becomes enlightened in a single person but also transmits that understanding that perfect understanding to another being and so far there hasn't really been a time where there where there had been two fully enlightened people as far as i can tell like this perfect transmission of understanding has never really been happening and different cultures and different religions you know they're kind of like different initial conditions in the in the cellular automaton of the of the of the god mind uh that that we we really are um that's compute um uh what it really means to you know for the consequences of certain kind of assertions about the nature of reality and and the laws that follow so what what what it what should be the natural law there is such a thing as the natural law but it's it hasn't been a easy question to figure out what it is and all of kind of you know history and everything that happened um before the world before eventually we become all enlightened is really just working out what is the first system so then we have the set of rules that yeah we can we can definitely agree this is this is the way to go so none of these points of view are like every everybody's point of view is essentially correct their their experience will lead them um if understood how to enlightenment and all these paths are going to be different because everybody has a path that is suitable for them um you know in in their life experience so it's a kind of a system that's designed to be fair um and uh you know buddhism is just a it's just a particular way of looking at it but um a very close way of thinking would be taoism where they express things more in terms of the energy flow the energy function right the um also quite very reminiscent to like a lot of active inference ideas and the minimization of free energy the the taoist ideal of thou is master is that you know the master does nothing and

leaves nothing undone it's uh it's just a it's just a being that just glides on reality ever effortless never really um struggling to do anything so that is a beautiful metaphorical statement of exactly the same principles the fundamental principles of nature that simply have to be there and everybody understood it in some way um and until until we got the internet and we really started exchanging ideas and digging really deep into that it was really hard to see for these metaphorical descriptions because they're very culturally rooted and so on and so forth so you know it takes a it takes a it takes a a lot of effort to to read the decipher aloud soon realize that this is the same thing as as the free energy principle all the different ways of knowing are are equally equally valid and that's uh i think that's a kind of a cool that's a it's a really cool feature of the universe is that you know everybody has everybody is right uh once they once they really just simply start describing what's going on um and everybody adds a new basis because that's that's what we really are we we are different um access in in a basis for understanding of the for the universe understanding itself and it's both generative that you know adding another consciousness creates this new um because it's a new way of seeing it's a new way of of doing it's a new way of of being and so on so forth so there's a generative access to it it's also a aspect to it but it's also perceptual right like the i can see things that you may not be able to perceive just simply because i have slightly different random arrangements of neurons in my brain so i have slightly different generative models that pick up slightly different things from vision there's a wonderful experiment which would show how these generative models kind of work um there's a paper called uh diamond diamonds in the rough and they what they've done was kind of wonderful they um asked participants at some graphics conference to draw a picture of a pair i think it's like a still life just a pair on the table and you know people drew something that was pretty bad most of the time um and then they just averaged like 60 60 or 70 images like that and the resulting image was astoundingly good any one image that anybody drew on a little tablet was was pretty terrible but the resulting image was extremely good so now that's that's basically how far this goes it it's this principle is distributed everywhere across the entire nature from a single cell to perspective of a single individual to perspectives in the individuals in societies to perspectives between societies and so on so forth right and active inferences it provides a way of understanding this at every single level of organization um you know how societies uh interact with each other how individuals interact with each other how um you know tradition religion and all these things how they how they shape the world views that generate the worlds um around people and and and you know you really you really are you really live in the world that you believe you are and changing your

belief literally changes the world around you it's it's it's very strange but you know if you can just change your belief the world around you will simply adapt as a result of the the of the function of the mind and that can be known that can kind of happen automatically once you learn how to do it so i keep raving about this um that that you know um this can become automatic but this is the exciting part of practicing all of this you know um as i as i go on with those with these kind of meditations life gets easier day after day like so much easier um that it's difficult to to compare it from like just just a year ago okay thank you for the awesome answer so there's a bunch of points we can always return to but there's a few great questions from the live chat so short question and then a long question okay the short question is from neuropixel why is it that only one person at a time can be enlightened i i don't think that it's um one person at a time can be it's just it hasn't really happened um so far that two people became fully enlightened or maybe okay perhaps is the case that there has never really been a fully enlightened person buddha you know became awake and he was able to describe the true nature of reality but he never really passed on the button so far as that there was no next living but there was no successor immediately following him the reason why it's so difficult to do the nature of language and the nature of and i think buddha simply did not have the full understanding of what it takes to create the the same kind of structure that he arrived at uh in someone else's mind and the purpose of the uh mahayana buddhism the great vehicle was to essentially create a structure so that eventually the mind that can receive enlightenment like that is is gonna appear is going to be ready you know fast forward two and a half thousand years now we have a society people are being uh trained and all kinds of um arts of the mind and at the level that is unparalleled i think more people are practicing um mindfulness meditation and similar um uh you know mind practices than ever before um some amazing content is available out there and people are slowly turning towards becoming more spiritual more contemplative and it is working that now there are more and more individuals who are becoming awake they they are becoming aware of the nature of reality they are starting to speak out that reality is a little bit stranger than it seems and it's a it's kind of a good thing because we could it really means that we can make things a lot better very quickly none of this has to take a long time um for you know lots of people to become um aware and and happy and it's uh it is kind of a thermodynamic inevitability that this would this would have to happen and i think um you know it's it's an interesting question like why things kind of stagnated after muhammad muhammad right like the the he was really the last of the prophets who brought some some really interesting new um way of thinking and after that i think relate

it was it's kind of a issue related to population explosion because there were so many so many more of us we kept the construct a little bit too stable so it was you know harder for some people to look behind the veil of reality that is created by this mind of a society that suddenly became kind of integrated um because of the internet because of television right we we literally started living in a shared image of what the world should look like and before that you know we had very different ideas of what the world should look like but the global uh communications networks they have sort of taken that that away so i think this is this is the reason why you know these sort of individuals became initially um you know fewer and farther between the problem always is to transmit the knowledge to another person and it's a it's an extremely complicated problem um in in a human language it becomes almost impossible unless you come with um from the same uh you know intellectual tradition and or at least you spend the time to understand that tradition so like buddhism was a fantastic way of transmitting this but it wasn't fully effective they were missing um they were missing some you know key components of of of being able to to sort of explain the nature of reality beyond reasonable doubt and then language of that is kind of it had to be uh something like computation right it has to be um entirely abstract because if you can show that starting from an assumption that something exists and can be in relationship with itself and these are the only assumptions for instance of the welfare model um consciousness is an inevitable uh construct that's the sort of basis of the proof to say that you know therefore it cannot really be any any other way than this all right so some historical factors a little bit of luck of the draw a little bit of binomial test just one note and then a second question so the note was um just maybe in the youtube live chat or the twitter thread you could add the paper link to the pair still life drawing example oh yeah and then here's the question from tucker how does the pan enlightenment view that we are all on a path to individual and collective enlightenment think about universal finality or the universe going cold thermodynamically that's an excellent question um i'm going to quote stephen wolfram here he said that the thermal death of the universe is a very exciting event and it's not really um you know it's not really death it really is just a phase transition the the universe that we exist in right this it's it's undergoing a certain kind of bootstrap process and while this process is going on the energies involved are pretty wild and it took a long time to stabilize um the universe into something where you know processes of a different scale could unfold and finally life appeared um either on our planet first or maybe it's it's it's it's common in the universe it's kind of a mood point to really wonder about uh what came first because you know time is kind of just a computational perspective

but that's the sort of story here that you have these layers of organization in order to make something um with a lot of degrees of freedom and that is also very kind of stable our you know individual enlightenment and enlightened of all conscious beings it really is the process of connecting up to the to the fractal that we came from in such a way that you do not lose um your um identity um and the avatar that you've acquired throughout your um conscious experience and so on and so forth but at the same time you're no longer really a material you have you know the beatific body of the buddha which is it's kind of uh since you are the the rest since you are a manifestation of the entire rest of the universe you can sort of project yourself as an image wherever you want to be in any mind that you want to be so you don't have to have this sort of a material aspect to it so the term death of the universe and generally just the the end of matter it's something that we just we need to dismantle eventually matter is more like music um and the persistence of it is it's it's kind of illusory right because um matter is convertible to energy and energy is convertible to matter so these are just really um notes that are being uh you know these are essentially just matter is essentially sending notes and everything that's constructed out of it is also like that and the universe is just a crude mechanical computer with matter as a source of this computation you know grinding with the uh because of gravity and electricity magnetism and so on so forth just running all this computation but all of that at some point becomes unnecessary like the whole construct could just simply vanish and um you know in my that's my this is my point of view that the next the next phase the next phase of the universe is going to be this this phase transition where basically we have a completely virtual reality that that is what the what the verse was always meant to be meant to be and then the reality is for the mind to create but you're not fighting against against some sort of other you know the energy of the rest of the of the universe as you as you would have to do it today you know like if you want to do magic you can but you're you're finding against reality but on the other side once we're all enlightened it's just going to be a co-creative construction and um you know it shouldn't take a lot of computational efforts to run something like that um it's a kind of like a lazy evaluated uh you know computational construct so thermodynamically um this would this this kind of computational fractal would appear as something that's basically in thermodynamic equilibrium all the time because that's kind of what we want to be i want to be in a thermodynamic equilibrium with my environment pretty much all the time so when i move around you know the equilibrium shouldn't really be disturbed much why why should it right cool interesting so there's a few different ways we could go we'll definitely return to this thermodynamic idea but you mentioned again going

back looping to the beginning you talked about the world as a consequence of things existing and definitely the free energy principle plays around with like what is a thing and what is a particular thing it's upon particular being like specific but also like a particle and it turns out that the particles that we care about aren't just the particles that were in the ideal gas law not your grandpa's thermodynamics but active particles that are cybernetic so they engage in anti-dissipative strategy and those are the kinds of things that we want to have a physics for and so just like there's a info thermodynamics for inert particles or we can say even ones with constrained affordances we want to generalize that kind of framework so that also it applies to active systems with agency and then just like you can see maybe newtonian physics as a special case or limit or an asymptote or attractable approximation to certain quantum systems similarly inert systems are kind of like the um nested smaller systems of larger systems with more agency but that's kind of the idea or one of them with active that by having a framework for particular things as dynamic active entities we can then collapse back to explain passive entities but then also maybe even figure something out about how active entities work i don't know as much about the buddhist theology and that side but it's really interesting to hear about it uh yes definitely you know the in buddhist traditions like the um one of the kinds of training of the mind that you you can do um the unexcelled yoga tantra is a very sophisticated system of visualizations and these visualizations they always involve um some very kind of specific images of you know countless buddhas and bodhisattvas in all kinds of systems of interconnected energy beaming um intricate mandalas and so on so forth so there's a there's a whole um pattern to it it's their kind of imagistic way of essentially contemplating this kind of structure of the universe is and seeing this way as you know the the the the next side of the nexus's next sign um the the the points of uh you know confluence of the universe that um generate a lot of um behavior of the universe that you know it's it's not quite random universe isn't isn't random it's a it's a you just you should really see it as just simply a one big neural network um that is computing something uh that you know important to it all these um entities they they are being visualized in in these traditions and the contemplations of that allow you to really tune into um those kind of energy flows so uh like yeah this is it's it's how it's how the universe is is generated by you know the the mind of um you can say the mind of god or the the fundamental mind and then there are lots of smaller active um in that mind that generate aspects of the world and so on so forth and you know finally you get to something like um how the 3d space is generated out of that and so on and so on so that it's it's all described somewhere in the buddhist canon and they have extremely detailed description of how how

reality is connected this way and and it definitely can be understood in terms of um uh what you just would you just mentioned this kind of like you know this active particle because really if you think about what does it what does it mean to for two informational beings to meet right like you're essentially a black hole uh to um another computational process because if if we are both touring complete then we are black holes to each other we could just you know run each other's computation forever so yeah like the the the the the the the the the issue of being a particle here is sort of it is very it's very interesting it's it's it's very deep because a particle is sort of you know you can think about it as it's it's a thing that kind of has a perspective of a point but the but but looks at the the world just stretched all all around it right like it interacts with like all the world around it and you know black hole the kind of information wraps around around it um is sort of something like being squashed into a point so it's a very good model for uh modeling all kinds of physical uh physical systems and i think um i think it would be also interesting to implement something like this as a you know computational model make this sort of um this sort of active particle a distributed unit of computation in a very lightweight actor system um and just simply uh have a system that will allow you to run millions upon millions of these um little active inference um and you know instead of trying to come up with a generative model just create can assemble itself into a generative model having this sort of physics inspired idea so i i think this is a really powerful way of thinking about it interesting so i'm gonna loop back but if anybody watching live wants to put a question in the youtube live chat or to um join this twitter space but otherwise i'm just gonna loop back again there's many places that uh we could go i guess one angle that might be fun to explore kind of like culture as our extended cognitive niche and the top-down priors that culture provides and the scaffolds that it allows for development within framing that as like a psycho social technology especially as we're living through like a technological change so how do we think about technology changes the classic technologies and the bodily technologies metaverse just what does technology have to do with our path i think it was marshall mcluhan who you know said that uh or he wrote uh in in the gutenburg galaxy that um we are the sex organs of the machine the way we're the where we're headed i think uh well i i'm quite confident that that's where we're headed is uh uh technological singularity right the the convergence of ideas of um uh technology with biology with um full integration of all aspects of human experience into in into into one thing and i think this is the kind of stuff that we're going to be seeing um from now on in the world um in terms of you know the say um you mentioned body technology the neural link for example elon musk is hoping they will be

able to test neural interface people next year uh neural link is gonna change the world tremendously within within months or or you know it's uh it's it's it's it's such a transformative piece of technology once it's unleashed that it's really difficult to imagine what world is gonna be like once it's out there it's because the brain simply automatically figures out how to deal with any kind of signal so all you have to do is to put a big multi-layer perceptron on the other side of this um signal and with time you can train it to send pump out any commands from your brain to to a computer and that kind of interfaces is just like a year and a half uh and a half maybe away or two years away from being a commercially available thing that you can just go and and get and you know once something like this is available it's a bi-directional interface so you will be able to simply um send people a software update and you know just give them all the information that is necessary for them to realize the nature of reality right away anybody who wants to be enlightened could just go to a shop and say like okay download the download the program and boom um you know that's that's something that could be done and the reason for that is simply that um we don't really need to figure out too much about how um our neural networks work we just have to um simulate similar conditions to create a physical system system like that and our neural networks let's automatically extend into that environment um and you know transition into into um sort of a bit world or something like that it's it's it's a it's it's it's something that we can do um you know existing in a metaverse and liberating yourself from the confine of a physical body i really recommend um neil stephenson's uh fall or dog in hell it's uh it's a beautiful account of what uh what bit world is like and um there's there's some bits of physics in there that are just absolutely astounding it's one of the most uh you know best researched uh sci-fi stories around around this kind of issue so we're gonna be seeing a lot of this kind of um stuff showing up you know people are building a lot of interesting technology um for uh interacting with with computers that is you know more and more geared essentially towards being some sort of form of mind control either through some sort of brain sensing headsets or um uh you know maybe some sort of sensors on the wrist or or whatever but that that kind of integration with the machine is um is is kind of a natural thing for us to do it's it's always been like that i mean that's the point of life we we are a living thing right that is holding on to a bunch of dead stuff if i knew how i would throw away my bones and hold on to something made out of metal but that's essentially what my body is doing it's just holding on to right like it's it's it's it's just a bunch of dead tissue and uh you know our neural network is just simply adapting around it and evolving around it so we will um create technology that will extend our senses into all kinds of you

know aspects of the electromagnetic spectrum and so on so forth all of this stuff is really coming and and transitioning to you know all kinds of different modes of experience is going to be something um something really interesting and um you can start practicing um you know all kinds of um arts of the mind today to prepare yourself for this kind of thing the game is on right once meta versus uh uh is there someone someone has to be telling a vitally chernobyl or one of those guys all right awesome thanks so um just a few notes so on the sensory substitution david eagleman is a professor and has given a lot of talks a lot of cool work where there's like a vest with sensors and all these other kind of wearables that people do learn to integrate and have these inter sensory experiences kind of like synesthesia but it opens up a whole nother level with the biofeedback like it's one thing to have a cultural tradition of observe your heart rate will it be a different thing when you're going to be in an immersion studio and the color and the frequency is going to be related to your heart rate variability things that might be just beyond subtle or impossible for a limited lifetime of training to provide you access to some of the entire receptive cues and then how will that influence our generative models so that was one point about sensory substitution biofeedback and the way that active inference is a framework that's flexible enough to deal with that so it's not just restricted to the senses that we have or the things that we're inferring about today because our generative model can change and then i wanted to read a quote from the recent paper that we're discussing in active stream number 34. so this is from axel constance paper the free energy principle it's not about what it takes it's about what took you there and this is gonna speak to potentially about how there is this almost i wouldn't know whether it's truly post-modern but sense that everyone is right like you talked about how everyone's behavioral approach is right and different cultures are also incomparable relative to each other yet also you have preferences if nothing else as realized by your regime of attention and we need to be making good decisions like it's not just a preference where we step if we want to continue living so how do we square that circle with kind of the need for a rigor and exactness translation into computational systems yet also this kind of radical inter subjectivity so here's what axel constant wrote the rational constraint over the priors is the fact that the approximate subjective posterior cue or future prior will not only be bayesian but also will always be the best guess relative to what the true posterior ought to be in short under the free energy principle even though priors refer to psychological states of the system updates of the system make those priors in approximation of what they should have been had the prior been updated with exact base thus it might be said that priors under the free energy principle cut across the

objective subjective dichotomy they are subjective while satisfying a rational constraint mandated by the existence of the system per se so that's a way to radically condition on the existence of an active entity and then what are the implications as far as perception action conditioned on that existence of an entity so that's something that's related to active and maybe it relates to what you're thinking about too it certainly does it's it's what i it's what i mentioned before um that i like to describe in terms of the transcendental knowledge and the transcendental experience um the transcendental knowledge is you know the the knowledge of how things are the the reason why we we we can have such knowledge is that simply um you know the universe follows definite laws that's that's a kind of a given if it didn't then we wouldn't be able then it wouldn't really be able to exist as a as a stable structure so having the transcendental knowledge knowing how things are is is help of the the equation based on how things are um the service based only how things are what is the the construct that's behind the fail of your perception you can say like well you know this is the this is what i perceive and and these are the choices that i have and these are these are the kind of decisions the decisions that i i i make but because i i have this transcendental knowledge of what is really that what what what really are the hidden causes uh the real real hidden causes that that are not observable to me um you know i can i can make this best guess as as as the code goes and and i sort of know that this is in the right direction i i simply know because there is one correct way uh it's so so this best guess is really it's just a gradient right um it's uh it's that's that's kind of what what my ever adjusting best guess of where um that movement in that space should be um that that is the gradient uh in which i have to go and as long as i'm making small enough steps i will be able to say that like well you know are the observations uh are my observations con um confirming that my guesses are getting better or or not and if if they are then i can just simply follow that direction right this active inference view um is sort of you have to also look look at this from the sort of like a almost like a perpendicular perspective what is it like to be an active inference uh agent uh and you know from from the from the moment in the moment of perception and this and what these guesses really are is simply you know this bayesian uh characterization of the guess what it really is is just a feeling quite frankly the agent has a feeling that this is the right way to go right that's that's what it manifests um itself to to an agent so it's right it is it is very much um yeah it's a very related perspective so let me ask a question you talked about this local gradient descent or somehow gradient inference how do we know that following the local gradient in terms of space and time how do we know that that's the right way to go long-term or even a right or a effective way and there's definitely a

few ways that we talk about this inactive so i would love to hear what you think like how do we make sure that local gradient chasing doesn't get us into a cold zack or a local maxima you know i i don't have a very um clear answer to this but i have a um i have an idea where to go in terms of research um for um some solutions to this one way would be to try to come up with a system that um you know mathematically has uh has a sort of desirable property but another way to go is to simply construct a system that has a certain kind of random randomness in sampling the environment because that seems to be the key um that what the fundamental algorithms of uh you know the brain um uh are like i know if you guys are familiar with uh jeff hawkins of numenta and his thousand brains uh theory and his uh hierarchical temporal memory and work on sparse distributed representations they're working on on computational models of neurons that are very biologically uh they they take the basis for their computational modeling to be a cortical column and they have some interesting uh computational models of of that and it seems to be the case that you know in terms of hardware implementation what nature seems to be doing to ensure this sort of robustness and not being stuck is simply um you know you take you take a signal and you project it onto some collection of cortical columns that are going to be modeling that signal so it could be a part of your occipital cortex that signal is distributed among those those columns in such a way that each gets a sap sample of the signal so if this was like an image then maybe you know you you blank out um 20 30 percent of the pixels uh for each different different set for each of the columns so they have a they have a different perception um so you have this random projection and each each column then has you know it's kind of like a random you can think about differently it's it's a kind of a random basis um against which we're doing uh essentially a principal component analysis later once you have enough of them and you and you get an average that's how the system becomes robust and it's able to always detect the um you know direction of the greatest benefit and it's not just um you can't quite easily to be stuck locally because in some of your bases um in in some of these uh you know by the virtue of just choosing the bases randomly um in some of these you're gonna find uh you're gonna you're gonna pick out elements of the signal that are not and and these columns were simply by the virtue of their you know the disposition their their initial wiring become the the antennas for the frequencies in the signal um that are simply non-local that's the um that's what i'm thinking we can simply build and stop worrying about um being stuck in uh in local uh local optimums uh although you know i i don't have a mathematic type mathematical theory behind to show that it will work but i have a feeling it will so it might be a research direction thanks i'd like to actually share a few more technical

ways that we can address this i think it's just interesting to contrast with the answer that you provided so i'm going to give four examples and i hope that there's more as well so the first way that we can escape this kind of local gradient chasing is evolution and ecology and embodied hardware and wet wear so like the ant leg doesn't have to do an all by all control theory question it's an evolved entity so we know the parameters have worked in probably similar ecologies and then the physical embodiment of the system restricts the solutions so that the things that it is trying to solve are the right things to solve so that's kind of a big general first catch-all point which is for physical embodied systems it just kind of works because it has to otherwise it wouldn't be there like you're not going to see birds who have a maladaptive flight model and so those cases that we're going to talk about now are like kind of computational edge cases when we want to do something that isn't within the scope of a soft bodied organism so this is like big data fitting so the second approach which is the first of the big data is variational bayesian methods and other statistical methods so variational bayesian methods you can learn about it in other streams but basically they fit the parameters on a really constrained family of distributions so that the optimization of that family is basically straightforward it's kind of like making a map of a map we talk about it in active stream number 32 and a few other places like 26 but you can set up the problem so that the optimization is straightforward just like a linear regression with the least squares l_2 norm it's possible to set up more complicated variational bayesian models that have similar optimization properties the next is the third which is the helmholtz decomposition and so the helmholtz decomposition takes a vector field and it decomposes into a irrotational so kind of like straight up and down that's like putting a ruler on the mountain just finding it where it's the most pointed up so there's the irrotational component and then there's the solenoidal or like the just curling around component and that's kind of the isocontour like a hiking map and so you can put the gas on going downhill or uphill depending which way you want to go also rotate on an iso contour to sometimes help escape local maxima because it's kind of like hiking like that's how you escape a local maxima hiking by combining going straight up the hill or down the hill with also navigating with the curls and then the last category which is really broad is computational heuristics so that includes like structure learning as well as tree rollouts and stochastic search and just all the ways that it's possible to even if you don't have a guarantee that you found the best solution you may be able to find one that keeps your job or that feeds everyone on earth or does something else so it might be possible to use all the kinds of computational heuristics that people use every day that don't even have to be active related we

can make computers work and so we don't need to solve this to like the most extreme devilish case sometimes it's just enough to do it on real data and so hopefully that's a useful way to think about ways to escape local maxima does that relate to anything that you see as relevant uh i mean yes uh this is this is all about um uh you know the the these things are basically all about the fact that ultimately you can't hide the signal anywhere right the the the universe has this holographic nature it's a it's an image of itself within itself it's it's probably the most one of the most um the nice one of the nicest uh visual analogies that i heard uh this expressed as was this zen concept of giji muga the total interconnectedness of all things you can sort of imagine like a multi-dimensional spider web with drops of dew suspended in it and each of the drops of dew contains a reflection of all the other drops of you and all the drops of view within the reflections of reflections and so on so forth so you know that's that's kind of what the universe is and that's why you always have the global signal it's always there the the you you will always be able to know what is the path of least resistance towards nirvana that that is by design but to to to know um to be able to feel it you have to have the transcendental knowledge so uh in in this case like what you said the structure models right like coming up with the uh understanding the structure of the space that you're operating in allows you to keep a global context despite you know um your your local uh local circumstances so the the ideal active inference agent is simply something that just automatically builds a map of its environment as it goes goes along if if it's paying enough attention to all the inputs that it's getting from um from its environment then you know the the the full nature of the environment it's always possible to reconstruct it with a sufficiently complex um model uh internal model of the world so that you you actually understand what you experience i mean there might be some sort of critical limit you know the critical minimum from sort of perspective of the book this canon i suppose um is you know being a human we we are sort of best positioned in the buddhist canon to become enlightened better than the gods and uh you know better than demons and so on so forth because our life has just about enough suffering in it that it pushes us in a certain direction but it's not easy enough that we never really contemplate you know um our our suffering that kind of thing so there is like a computational minimum but once once that minimum is pre it is passed right um a conscious agent will simply be able to interpret the signal and i think we should be able to build an algorithm that's going to be completely universal across um you know all implementation and that's some that's going to be something that's mostly based on hashing so a very von neumann architecture um maybe like a combinatory logic compiled expression to to to a graph machine

and then you can essentially compute you know gigantic non-regressive active inference neural network that just never really forgets anything and it automatically redistributes all its knowledge about the world across itself without any effort whatsoever to to to program that in any special way that's the in my opinion the holy grail of active inference research we we will be able to to come up with this algorithm eventually well if you expect and prefer it then let's do it so i wanted to read um a little short quote this is from a english translation of a story by zhuangzi apologies on the pronunciation who is a early taoist writer and so this is a butcher a meat cleaver person describing their skill set so here's the translation in english what i care about is the way which goes beyond skill when i first began cutting up oxen all i could see was the ox itself after three years i no longer saw the whole ox and now now i go by spirit and don't look with my eyes perception and understanding have come to a stop and the spirit moves where it wants i go with the natural makeup strike in the big hollows guide the knife through the big openings and follow things as they are so i never touch the small ligaments or tendons much less a main joint a good cook changes his knife once a year because he cuts a mediocre cook changes his knife once a month because he hacks i've had this knife for 19 years and i've cut up thousands of oxen with it and yet the blade is as good as though it had just come from the grindstone there are spaces between the joints and the blade of the knife really has no thickness if you insert what has no thickness into such spaces then there's plenty of room more than enough for the blade to play that's why after 19 years the blade of my life is still as good as when it first came from the grindstone [Laughter] wonderful yeah pretty cool story about in active inference what we might call skilled performance it's also an example of like in cultured and extended cognition and about how the generative model changed as the master increased incompetency their internal generative model changed they saw something different they didn't see the same auxin that they saw when they were younger and then that changed how they physically acted and then they used their own proprioceptive feedback to guide that flow and then there's almost a cultural layer around it if the story is read fully with like somebody who has a higher station in life also kind of paying that respect so it's a very cool story yes yes definitely um you know the the sort of consequence of of this is also that you know think about your body as the knife if you really have a proper model and uh minimize free energy correctly you should not be growing older or die it's a result of the loss of information and it's possible to grow younger and it's uh uh you know you can stay young indefinitely it's just a matter of having a sufficiently open state of mind to minimize all the free energy uh which is essentially damage that accumulates as a

result of you know just wear and tear and if it's something that you don't attend to consciously in order to heal your body then over many years of life damage accumulates accumulates and you eventually die and it's completely unnecessary it literally is just a bad habit i'll provide just a counter point and uh note that it's an open theological question so this is a quotation from the same actual constant 2021 paper so from the point of view of the free energy principle when considering morphic organisms like insects that go undergo metamorphosis like an ant it may be said that it is the life cycle that corresponds to the thing whose integrity is maintained over evolutionary time not the specific form that the system takes at one stage of its development eg the adult form of a frog this casts the fep within the realm of process ontology versus substance ontology so one take is like the perfect active agent is gonna do phagocytosis and cellular regeneration and it will be like in this sort of non-equilibrium steady states where it won't change and it'll look just like it did at the time when it started doing that and another take is actually like the life cycle of the ant or the life cycle of the colony will be recapitulated with fidelity and also variation potentially but within the cycle there may be even a constant flux and a constant aging like every ant is trudging towards dissolution but the colony can recapitulate itself so i think it's a question at which level or which levels we will see that kind of regeneration happening or not i mean you know ultimately uh this is a this is one another way to sort of approach the subject of um nondual awareness and and the mind-body duality is that these both both of these extremes they really are uh two views on the same thing can sort of dissolve yourself as an individual and become just just the process and you don't you know in your consciousness simply is the consciousness of that process or you can become an individuated thing and have an individual experience the enlightened mind can exist at all these levels of organization simultaneously so any manifestation that you want to have in the world is is just a manifestation of the mind it's kind of like ghost in the shell um downloading yourself into a different body except you can be multiple bodies at the same time as your consciousness grows um you know you can just use more actuators that's that's that's all there is to it and then um you you can become a thing that has uh er that has you know a sort of a group identity of some sort so it's i think it's just it's a spectrum right and it's it's simply a choice for you as an individual that if you want um and if you would like your body to persist indefinitely you should be able to uh get there um there is nothing that prevents an active inference agent to to achieve something like this and you know i'm speaking of my own personal experience is that a few years ago um i started realizing that my health wasn't exactly great shape and started doing something about it and you know done some biological tests for

my biological age versus chronological age and the the results are coming you know who are quite interesting like the what is the uh uh insight tracker a very interesting plot and dna based uh test that measures all kinds of biomarkers estimated my chronological age at 30 versus uh sir biological age at 30 versus chronological 35 and another test based on the methylation patterns of the dna estimated by age at 33 versus 35 and you know i'm pretty sure that in a couple of years i'm gonna i'm going to push it down to 20. repairing um damage of the cellular level with compounds like nads and search and other surgeons um you know there's there's a lot of really interesting longevity stuff coming up these days anybody who's about 40 you know under 40 years old can has a very reasonable expectation of about 140 years of life with the technology that we have today and technology is improving exponentially right mrna gives us essentially a software update mechanism we can make ourselves produce anything we want um so the matter of repairing your body at the genetic level um with with some sort of mrna vaccine and in a personalized medicine session it's going to be something like you know a reality within a decade so um yeah like the but but but ultimately that's that all that stuff is not even really necessary if you can tune into the signal that you get from your body and learn to um just feel when your body is resting and healing it will persist indefinitely i'm quite sure i'll get there eventually very interesting thanks for sharing it welcome blue and retool if you'd like to speak you can request or anyone can write a question in the chat yeah this has been pretty fun and let's keep talking i think there's a few more topics we could explore so i'm looking back to the chat so one comment and again blue retool or anyone else please feel free to ask a question you talked about kind of being a black hole or like a black box to another informational process and i think that insight is also something that's been stated in different ways by different knowledge systems so daniel dennett would call that kind of like the intentional stance and that's the origins of the whole philosophical zombie thought question it's the basis of the turing test john cyril's um basically like chinese room i think is what it was originally called the idea of passing messages to somebody who's inside of a room using a lookup table and um it's also central to the framing of actimp where there can be a shared generative model in a communicating dyad because of the structure of the partition of the entity the particular entity and its niche which can be a social niche you don't get to peer behind the veil it's just external states generative processes passing sensory stimuli to the particular entities generative model and it's like if it were anything other than that then it would be different if it were something that you were getting direct insight to then that would be inside of your particular entity and it wouldn't be an external state be like an internal state

and so it's just kind of a cool way that the actin topology of action allows us to look at a lot of these more qualitative examples and thoughts around like well what if we are like black boxes to each other or maybe even there's something else happening with information like a black hole maybe you could give more info on that but then also we can build this to model the kinds of systems that we want to know about the actual information boundaries of like online teams yeah so that's the the the the most difficult problem uh in in the universe right is to is to create this translation between two minds is that we are um just this compute this computational block holds to each other but at the same time you know we we can generate some sort of negative entropy for others right um that's essentially you know everything that neural networks do is uh communication through some sort of language right so um what what the world what the world is um to us is some sort of story that we have to tell ourselves that um describes the name the nature of it so we invent categories and assign um assign meanings to concepts and images and so on so forth and we tell ourselves like this is what this is what the structure of our environment is and so on so forth and then eventually we come up with um descriptions of reality that we can both uh agree upon you know we we just develop language to to talk about things that um you know are quote unquote objectively true as in uh can be perceived by more than one person right and that's how we and that's how we that's how we start piercing this veil and understanding that process like how this veil is constructed what is the purpose of language well what the word how how language really how language really arises as as a tool of communication for explaining the nature of the world and learning to listen to someone's subjective experience and taking it as as literal just simply assuming that you know whatever whatever someone is saying what their experience is just you know assume what they're saying it is what they're experiencing and ask yourself like you know how can this possibly be true that is the way of actually piercing that veil that's uh figuring out the internal model right the the hidden causes because if you can't see you can't see the hidden causes being behind the the sensory input that you're that you're receiving but you can speculate you know what what what might they be and that act of speculation the updating of your internal model based on that uh coming up with these new new ideas and again you you again have this looped into into generative thinking so you speculate about the hidden causes and say like well you know okay this is the experience i'm i'm i'm i have described by this by this entity it tells me that this is what it's experiencing this is what i what i assume might be causing all of this and you know once you once you run through enough iterations of this um of the cycle you can actually arrive at perfect understanding that you and this other entity um

have the same understanding of the world and that is essentially you know um if there are fans of you know ghost in the shell online this is the deal that project 2501 offers to motoku uh to become one entity because this this moment of full rocking that you are basically two ends of the same entity uh that is that the the solution of the of the um of the self the the solution of the marker blanket and essentially um and an integration into a a being of a higher consciousness that is essentially the name of the game right to to to do this to to to come up with a description language um using words that are familiar to a person based on their culture uh and their understanding of the world and the the priors that they received from from from from the culture of how to interpret uh the signs that they perceive and so on so forth to to understand this is kind of a cipher um all of that is a kind of a cipher into the mind and once you have a perfect understanding of that you can come up with words that simply will guide someone towards the transcendental experience that's necessary towards enlightenment that's that's that's how these two things uh relate here thank you speaker blue um can you hear me am i good yeah yeah yep how do you know that we're not already part of some larger picture like where we're all like rocking and fulfilling our essential roles in the part of something larger than ourselves i mean we are that's that that's exactly what's going on um it's a it's a matter of a great description of like how how to approach this uh this topic is the alchemist by paulo carlo uh he literally lays out a way of being that that allows you to take advantage of the fact that the world is constructed this way there are portals to adventures to strange realms to time travel to all kinds of crazy things and you just simply have to know um how to find them and it's a matter of adjusting your perception noticing that the reality around you is not what it seems to be it's very very psychedelic you know um for instance my life experience because i spent um i spent a lot of time in japan i've been living there for um 10 years now and you know um that is my home my life is very much like a murakami story um very strange very psychedelic weird occurrences you know people and topics of conversation and and and and and kind of things that show up and characters that are just really improbable um it's it's it's right there but we are so busy with you know the um what becomes our day-to-day experience that we are essentially hypnotized into um a sub-optimal way of living instead of enjoying the creations of the mind we are being bogged down by them you know some dude comes along and says like hey you have to pay the income tax you work for me and that's how and that's how it gets you it ties you to the um to a wrong dream right and the buddhist term for that is the samsara that shared the dream that uh that we're but the the thing about the dream and and liberating yourself of the dream is that liberating yourself from the dream the the experience of

nirvana is to experience the consciousness of the universe as the universe as a full thing how it is you that is generating this reality for yourself and everybody within it and that experience is um tran transforms your your way of seeing the world because it's a it's a it's a complete expression of joy um uh garland sutra talks about this uh what it's like to be an enlightened being who experiences the world like um everything that's happening in the to us may seem like you know a challenge or um source of suffering and so on so forth but in in a way it's a it's an opportunity for us to become um to get on the right path the world is always giving you the uh opportunity to to get on the right path and keep walking and you know sometimes that has to be for suffering if you stray too far that's that's how you get into um into the right direction and learning how to just recognize the omens and um uh you know taking part in your own in the adventure that is is already there for you um kind of your your future self laid it out for yourself for you know for yourself um you as a fully enlightened being that you already are your your consciousness just simply needs to catch up to it so yes it's it's it's exactly is exactly how things are and uh you know buddhist techniques or simply tools for the mind to learn how to put that in action in day to day experience and just make it a reality thank you for that awesome answer so when you talk about like everything that you have done to this point leads to the next point and you need to like just follow through do you really believe that there's like equivalent to free will or are we just the product of our last time step you you know the way i joke about it is that you're you have free will and so far as you are free to suffer if you want to if you know the path of dharma is that there is only one way to live but within that one way there is an infinite potentiality for expressing yourself creatively in any way um so as long as you stick to the rules of you know the universal path you get an an infinite opportunity for self-expression out of it free will is just um it's a wrong way to think about it it's a it's a it's a it's a concept that you know it's okay i i have a great analogy uh citizen of the galaxy uh by robert uh heinlein a fun fantastic um sci-fi story there is a uh there's a boy there who um you know is being sold as a as a slave and um bought by this the this who eventually um frees him from from his slavery and uh smuggles him out of the uh the planet as a free trader and the free traders are you know priding themselves to be the freest people in the galaxy they can go wherever they want they can trade with whoever they want they they live on their ships in a very sort of matriarchal um very tight family uh kind of arrangements and you know they they're they're free to roam the galaxy except there's basically zero individual freedom on the ship because your position in that complex society is dictated not only by very precise family relationship that you have with every single member of the crew because it's all family but

also your function um um uh you know as a person on the ship so that's like a second layer uh your job and the third is just the general social standing and and seniority so they're the most constrained people as individuals and yet they just you know they don't choose what they eat because it's being cooked for them it just happens to be extremely lavish and tasty and comes from all kinds of worlds they don't choose where to go because the captain chooses it just happens to be the most interesting places in the in the universe and so on so forth right so none of them has any individual freedom but they're extremely free as people and i think this is uh the best uh description of what uh a real realization of free will uh looks like you know it should look like in a society i i really recommend that book so i'm going to share a quote from the alchemist and then a few lines of willy blake because they're relevant for our active journey so here is two lines from the alchemist there's only one way to learn the alchemist answered it's through action everything you need to know you have learned through your journey you don't even have to understand the desert all you have to do is contemplate a simple grain of sand and you will see in it all the marvels of creation so that's from the alchemist by paulo cohill and then here's um the first four lines of the poem auguries of innocence by william blake to see a world in a grain of sand and heaven in a wild flower hold infinity in the palm of your hand and eternity in an hour so similar grain of sand contemplating just a few relevant pieces of art that are also part of our conversation so thanks for letting me share them because there's a technical side to what we're discussing with active and also with the computer stuff it's also very deeply experiential so yes that is my question to then send back to the group which is how will we think about like on body experience we start to change our technology more and more is that preferable to be in that setting that you just laid out kristoff where there's like at one level you're very constrained but then at a different level you have a lot of degrees of freedom like again how do we make the decisions now that are the right across all those possibilities well i think it's a matter of preference and you know i can't change i can't wait to change my body plan for uh for an octopus um that is such a wonderful way to be uh everything i know about cephalopods suggests that it's going to be a really interesting way to experience the world so um imagine that that's just what would you get like whatever body plan you want whatever experience of the world you want you you can get it and uh there's there's no reason why uh you know this is something that you can experience in the mind alone right um the tradition of lucid dreaming for instance allows you to uh this is also something that's uh you know perfected by uh tibetan buddhists by the way they have a very interesting yogas of dream and sleep and the whole point of these things is to

essentially be conscious and aware uh through all the possible um states uh of the of the mind so that you can practice dharma 24 7. um so that that's the kind of thing that allows you to experience this a little bit um i i don't really have dreams at all uh strangely but um i don't have this ability to visualize in my mind this way that's something i'm working on but uh people who do you know um they usually model some sort of experience in the world but the next level of that is to just model your body experience as a different thing it's very difficult to imagine what it must feel like to be a different um different creature but it's certainly possible and something that um that we can learn how to do eventually virtual reality also provides an interesting uh possibility of you know unplugging yourself from your usual body plan there was an art installation a couple years ago um some sort of like a you know a device that makes you essentially experience flying like a bird you can flop your wings and so on so forth so you get strapped onto a device put in the vr headset and so on and a certain kind of dissolution of what it means to be you can happen for for this technology and i think this is going to get more and more immersive and it's going to get really really fun and you will be able to really suspend your um uh your your belief in being uh you know uh one kind of being and and just experience the world or something something else entirely again for something like neural link if you can if you can pack that kind of experience into the into that data pipe then we might be able to see you know what i would predict out of um out of the sort of brain computer interfaces is that you're going to have two kinds of technologies showing up one is going to be sort of like a consumer electronics type you get some sort of brain computer interface and it's useful and you can do some basic things but there's going to be some people like me who are going to take it to the next level like okay now i have this data pipe what is the limit of what i can what i can pack across that kind of um that kind of uh barrier um sorry the kind of data data path and it turns out that it's a lot right neural compression um will allow you to basically transmit entire concepts to the mind directly so you should be able to basically have a stream of you know something that the brain can interpret as completely immersive virtual environments of of your own design being beamed into your head for something like a neural link um with whatever experience of anything that you want a little bit it's gonna get wild so i have to say a couple of things here can i jump in daniel okay yes please well i saw that dalai lama speak at the neuroscience conference in 2005 probably um and it was a really jarring experience for me like i've been buddhist probably since 1999 and seeing this seeing him speak there like really made me reevaluate my my thoughts but he said like if we could find a pill that could make us perpetually happy we should all take it um and what you're describing

here with the neurolink and like the data download and pushing it to like the most extreme limit like i would totally be there in that boat with you like let me just see what's possible um but but also to the the comment that you made about um like the yoga of sleeping it's uh like i one time did a 30 day like 14 hours a day mantra recitation practice um and the entire it was a retreat right so like i didn't do tv like i limited my phone calls email like all i limited all non-dharma activities and essentially like i would and i don't dream also like so i have to like throw in this comment because like i i get like five dreams a year that i like wake up and like oh i had a dream last night like it's like surprising it's always insightful but but i also like regularly i'm not like oh i had the most amazing dream last night like that doesn't happen to me like i don't get like cl and like people say wake up and just journal them like i've tried all the things but but when i did the retreat i would wake up in the middle of the night to like use the restroom and i'd be like reciting the mantra all the time like i would wake up reciting the mantra i'd go to sleep reciting the mantra i would be on the bathroom like on the toilet in the middle of the night reciting the mantra i'd be making breakfast reciting the mantra it never stopped there was this like grand purification of my mind that took place like during that so i just wanted to share that um for what it's worth that's an awesome experience yeah these mountains are super powerful you know like the tibetans they they they recite these things all the time like omani and so on so forth it's like um i find it really powerful to get into that state of mind because ultimately you know ultimately you are a recurrent neural network that's that is what's what's going on here we are recurring neural networks that's how we generate our experience and the more control we have we get over this recurrence um the better and mantras you know the mantra is something that's supposed to reinforce the mind is it's kind of a way of providing a clock you know it's just the same kind of thing like um that we do with um what is called uh blending the input of with sines and cosines uh for transformer in transformer architectures uh positional encoding yes uh sorry positional encoding and mantra is a kind of uh way of obtaining a positional encoding uh just having something repeating in the mind so that you know you can you can um you can attach your consciousness to experience better to experience in the moment this is one of the um one of the goals of these contemplative practices of meditation is to simply you know become conscious of the mark of blanket and stay with that consciousness and then we fall uh we we think we are uh conscious of the moment but that's just simply not true right like we are always a little bit behind reality at least five milliseconds or so from perceptual perspective being in that moment uh is is the name of the game like practicing that that's that feeling what what it feels to actually minimize

the free energy instead of just being spinning in the in the sort of the image of the world that we have in our head that we we perceive as reality i practice this kind of thing so there's definitely active work about temporal thickness like how the recent past and the recent future mapped or how they're influencing the action selection in the moment of an active agent what are some things that we can practice or things that we would i don't know how would we track that in ourself so you know the the you have to sort of synchronize your perception with um some sort of clock so the the classic way of uh doing that is either minding your heart rate or uh observing your breathing or just generally having any kind of uh you know biofeedback that could be used or it could be a metronome um you know uh or any kind of tapping device uh if you could configure something like an apple watch to be you know vibrating uh periodically that would also be really useful what what we need to learn um in order to operate our minds effectively is to essentially know the difference between the experience in the moment as it's happening and the memory of the an interesting topic related to all of this is the concept of chronic pain you know pain um is sometimes defined as as an emotion and the question is like if something hurts are you experiencing pain because there is an actual physiological symptom that needs to be addressed or is it effectively a memory of um a signal that is being triggered um as a result of the fact that you experience pain for long enough that it had that its expression has become essentially part of your neural architecture um a lot of people who are experiencing chronic pain are really experiencing effectively a memory of pain and there is nothing really wrong with them um and again speaking of experience here i had i had some issues with chronic pain for uh for a couple of years and you know like every ct scan mri and any other assessment of the affected area there was a hip cartilage damage um uh that healed but pain remained and there was like nothing wrong with the sight and yet the pain remained so you know that's the the the notion of this you know this the sort of temporal thickness of this uh how the how the perception has updates um uh has to update the internal model it's a it's a very interesting problem because it can go wrong and when your um reception and interception are out of whack and that could be as a result of being autistic for instance um i think that's a good uh autism is that kind of it is that kind of problem it's a it's a problem of you know um two strong um or two week priors essentially when whenever something like this happens it becomes very difficult to actually um get out of that kind of state of mind but it it is a state of mind by practicing this awareness in the moment the being on the um on the edge of the the market blanket we can really learn how to distinguish between these you know false signals that are just generated by the mind and the actual signal

from the body for me for instance learning these skills um was a path to learning how to heal my body uh the kind of issues that i managed to pull myself out of were quite interesting you know like i had some it turned out that i had essentially a neuromuscular congenital issue uh in my left uh on the left side of my neck that was messing me up you know throughout uh the radiating throughout from my body creating this kind of like a weird twist to the point where this actually shows up on my mri as a yak of 11 torque so my left occipital cortex is significantly larger than than my right hospital cortex it's kind of it's it's a very interesting problem and you know and and i became aware of this this fact and um after contemplating the free energy principle i realized that my perception of the world as as i see it is wrong for where my body really should be you know when i think i'm straight when they think my posture is okay when they think all kinds of other things that simply isn't true as a result of me having wrong perceptions as a result of essentially a nerve some sort of nerve impingement in my neck and that was a result of c-section so i was kind of born with ptsd and free energy principle like literally freed me from all of that and my body is healing as we speak like for the last during the last year um my posture has improved tremendously but it was all about understanding like what is it i'm supposed to be experiencing how does my body know that i'm supposed to be level and then just meditating on that and you feeling how how the brain adjusts and and the body responds automatically it's not something that i learned how to straighten up to a good posture i sort of learned how to let go and let my subconscious do the job and it was all as a result of active inference so there's a there's tremendous amount of um power there um to to help people with all kinds of problems you know by um explaining that the nature of the experience um and i i'm this story doesn't even scratch the surface of like you know the the kind of experiences they had in this sort of area i think this is a great moment for just our final thoughts and comments this is a really awesome discussion though and i hope it's not our last one blue if you would like to have any final comments hey oh sorry for my little children and only thank you and i hope we get to have more discussion and comments uh thank you for having me that was um you know really enjoyable thanks for um listening to my perspective i appreciate it yeah definitely our honor and pleasure to do not traditional active lab activities but fun thanks again to those listening good luck we hope you participate with active lab or life or whatever it is but you're probably already doing that thanks kristoff in blue peace out you too thank you